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EN.705.640.8VL.FA24: *Cognitive and Behavioral Foundations for Artificial Intelligence*

Title: Evaluating Artificial and Alternative Intelligence: A New Framework for AI Evaluation

Executive Summary

The key question driving this research is: *How can a comprehensive evaluation framework be designed to account for both human-centric replication and superhuman capabilities in AI systems?* Traditional benchmarks, such as the Turing Test, have historically focused on assessing systems' ability to mimic human intelligence. However, this narrow focus, shaped by the definition of AI as human mimicry, restricts the scope of evaluation and fails to capture AI's broader potential to surpass human limitations in areas like large-scale data processing, scientific discovery, and adaptability to dynamic environments (Negrotti, 1991).

The proposed research introduces a comprehensive framework, the 'Multidimensional Intelligence Evaluation Framework' (MIEF), to address the inadequacies of traditional evaluation metrics by holistically assessing AI systems. By accounting for both human-centric replication and superhuman capabilities, the MIEF provides a balanced approach to evaluating AI systems that reflects their evolving roles in society. Unlike traditional anthropocentric benchmarks, the MIEF explicitly integrates superhuman capabilities, adaptability, ethical alignment, and sustainability alongside human-centric replication, offering a comprehensive tool for guiding AI development. This framework acknowledges the inherent trade-offs in evaluation, recognizing that an AI system may lack in one area but excel in another, ensuring a balanced and equitable assessment. By redefining AI evaluation to emphasize its role as an alternative intelligence, the MIEF promotes leveraging AI's unique strengths to aid and complement human capabilities. The impact of this research is twofold: providing a robust foundation for assessing AI systems' strengths and guiding their development toward ethically responsible and societally beneficial applications. Ultimately, this framework aims to shift the paradigm of AI evaluation, fostering innovation and ensuring alignment with long-term human and environmental values.

Topic Proposal, Literature Review & Thesis

The proposed topic examines how the evaluation of Artificial Intelligence (AI) systems changes depending on whether AI is defined as replicating human intelligence (Artificial Intelligence) or performing tasks beyond human capability (Alternate Intelligence) (Negrotti, 1991). The definition of AI fundamentally shapes its evaluation. When AI is narrowly defined as human mimicry, traditional benchmarks like the Turing Test dominate, assessing systems' ability to replicate human-like behaviors. However, such metrics fail to account for AI's transformative potential to solve problems at a scale and speed unattainable for humans. By redefining AI as an alternative intelligence, this exploration emphasizes its ability to complement and surpass human intelligence, addressing practical challenges and societal needs. The proposed multidimensional framework, MIEF, evaluates AI across both human-centric and superhuman dimensions, incorporating trade-offs to ensure a balanced assessment.

Massimo Negrotti's *"Alternative Intelligence"* critiques the long-standing ambition of Artificial Intelligence (AI) to replicate human cognition, arguing that this goal is both scientifically impractical and culturally rooted in industrial-era values. Negrotti suggests that AI's true potential lies not in emulating human thought but in developing unique, computational capabilities tailored to augment human problem-solving in ways beyond human limitations. By focusing on the intrinsic differences between human and machine intelligence, the paper challenges the foundational assumptions of AI research and advocates for a paradigm shift toward "Alternative Intelligence" (AltI). This shift redefines AI as a distinct, formal framework for intelligence that complements human abilities rather than imitates them, enabling systems to excel in areas such as processing vast datasets, high-speed computations, and scalable problem-solving. Negrotti's work highlights the need to rethink evaluation benchmarks for AI, moving away from human-centered metrics toward frameworks that celebrate the unique strengths of machines (1991).

In *"On Impact and Evaluation in Computational Creativity,"* Alison Pease and Simon Colton critique the reliance on the Turing Test for evaluating creativity in AI, arguing that its focus on imitation stifles innovation and fails to capture the essence of creative acts. The paper highlights the shortcomings of Turing-style evaluations, which reward mimicry and superficial traits, such as adherence to human styles, while penalizing originality and contextual relevance. To address this, the authors propose two alternative models: the FACE model, which evaluates creativity through generative acts (eg. framing, aesthetic measures, and conceptual development), and the IDEA model, which emphasizes the impact of creative acts on an ideal audience, incorporating factors such as well-being and cognitive effort. These models provide a flexible, process-oriented framework for assessing computational creativity, encouraging the development of systems that expand or transform conceptual spaces. By shifting the focus from human mimicry to machine-specific strengths and audience impact, the authors offer a progressive approach to evaluating and fostering innovation in creative AI systems (Pease & Colton, 2011).

Hannah Silva's *"Alternative Intelligence"* explores the intersection of human and artificial intelligences through the lens of personal narratives. Silva critiques the prevailing narrative of AI as a tool for replicating predictable, normative outputs, arguing instead for its potential to subvert and challenge dominant structures. She describes her own interactions with a language model (GPT-J) as a collaborative process that revealed unexpected, glitched, and poetic outputs, reflecting an "alternative intelligence" that defies the constraints of traditional AI paradigms. Drawing on neuroqueer and glitch feminist perspectives, Silva advocates for AI that prioritizes creativity, failure, and the transformation of human narratives rather than reinforcing normative patterns. This work challenges the reader to imagine AI not as an extension of human logic but as a partner in generating alternative ways of thinking and being (Silva, 2024).

Tian et al.'s *"Towards Human-Like and Transhuman Perception in AI 2.0"* explores the evolution of AI's perceptual capabilities, emphasizing their role as the interface between intelligent systems and the real world. The authors highlight six key research directions shaping the next generation of AI ("AI 2.0"), including advancements in human-like and transhuman active vision, auditory perception in realistic settings, natural speech perception, autonomous learning, large-scale perceptual data processing, and urban omnidirectional intelligent systems. By comparing current limitations, such as the disjointed development of algorithms for separate perceptual tasks, with the integrated approach of human sensory systems, the paper argues for intelligent perception systems that not only simulate but surpass human abilities. The authors propose a roadmap toward achieving these goals through innovations like active and multimodal learning, deep integration of perceptual modalities, and the creation of large-scale intelligent platforms, ultimately positioning AI as a transformative tool for applications like smart cities and autonomous systems (Tian et al., 2017).

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The paper *"Highly Accurate Protein Structure Prediction with AlphaFold"* by Jumper et al. presents a breakthrough in computational biology with the development of AlphaFold, a deep-learning-based system capable of predicting protein structures with near-experimental accuracy. Addressing the long-standing "protein folding problem," the authors highlight how AlphaFold integrates physical, geometric, and biological principles into its neural network design to predict three-dimensional protein structures based solely on amino acid sequences. Validated in the CASP14 competition, AlphaFold's predictions significantly outperformed existing methods, achieving a median accuracy of 0.96 Å compared to the next best method's 2.8 Å. This unprecedented level of precision enables the rapid and scalable determination of protein structures, solving problems that traditionally required years of experimental work. By demonstrating applications across structural bioinformatics, AlphaFold has set a new standard for predictive modeling, showcasing how AI can revolutionize domains where human capabilities alone face limitations (Jumper et al., 2021).

Korteling et al.'s *"Human-versus Artificial Intelligence"* examines the fundamental differences and complementarities between human and artificial intelligence. The paper critiques the anthropocentric tendency to use human intelligence as the gold standard for AI development, arguing that AI's strengths lie in its unique capabilities, such as rapid data processing and scalability, which often surpass human cognitive abilities. The authors advocate for a non-anthropocentric definition of intelligence, emphasizing the ability to achieve complex goals in varied environments rather than mirroring human cognition. They highlight the potential of narrow AI systems, integrated to complement human limitations, rather than pursuing human-like Artificial General Intelligence (AGI), which is both challenging and of limited practical value. The paper concludes with a call for "Intelligence Awareness", educational programs to help humans understand and effectively collaborate with AI systems, maximizing their complementary strengths (Korteling et al., 2021).

These articles collectively challenge the adequacy of traditional benchmarks, raising critical questions about how AI systems should be evaluated in light of their evolving capabilities. A central issue is how evaluation frameworks can effectively balance the competing paradigms of human-like replication and transhuman capabilities. Traditional metrics often fail to capture the unique strengths of AI systems, such as their capacity for rapid computation and large-scale problem-solving, while undervaluing monumental potential beyond human mimicry. Additionally, the articles highlight the need to incorporate dimensions such as creativity, adaptability, and societal impact into evaluation frameworks, recognizing that AI systems should not only mimic human intelligence but also innovate in ways that fundamentally reshape fields like science, art, and engineering. Addressing these challenges, the proposed thesis asserts that *"The evaluation of AI systems must adopt a multidimensional framework that accommodates both human-centric replication and superhuman capabilities, incorporating metrics such as creativity, adaptability, sustainability, ethical alignment, and value added, to foster innovation and align AI with societal needs."* This approach aims to redefine how AI is assessed, ensuring its development aligns with its potential to transform society.

Analysis

The evaluation of artificial intelligence systems requires a multidimensional approach that moves beyond traditional benchmarks, which focus primarily on human mimicry. The Multidimensional

Intelligence Evaluation Framework (MIEF) addresses this limitation by integrating dimensions such as human-centric replication, superhuman capability, adaptability, sustainability, ethical alignment, and value added. This framework ensures a holistic assessment of AI's role in society, recognizing both its ability to replicate human cognition and its capacity to surpass human limitations.

To operationalize the MIEF framework, each dimension is defined with clear metrics for evaluation. Human Mimicry assesses AI's ability to replicate human-specific cognitive and emotional behaviors, such as natural language understanding and emotional recognition. Superhuman Capability evaluates AI's performance in tasks that exceed human abilities, focusing on precision, speed, and scalability. Adaptability measures an AI system's flexibility in transitioning across tasks or adapting to dynamic environments. Value Added quantifies AI's contributions to productivity and innovation, highlighting its ability to achieve outcomes beyond human capability. Sustainability focuses on environmental and resource efficiency, using metrics like energy consumption and carbon footprint. Ethicality ensures alignment with societal values, emphasizing fairness, transparency, and accountability in decision-making. These metrics enable a structured evaluation of AI systems across diverse applications, providing a robust and holistic approach that transcends the limitations of traditional benchmarks.

AI evaluation has historically been limited by its anthropocentric definition, often prioritizing human mimicry over transformative potential. Massimo Negrotti critiques this focus, emphasizing that AI's true value lies in augmenting human problem-solving through unique computational capabilities (1991). For instance, systems like AlphaFold transcend human limitations by solving the protein folding problem, a task that would have taken decades, within hours (Jumper et al., 2021). Examples as such highlight the necessity of redefining AI as an alternative intelligence capable of complementing and surpassing human abilities.

Human intelligence, while remarkable, is fundamentally limited by ingrained cognitive biases and neural constraints. Over 200 cognitive biases have been documented, reflecting systematic distortions in information processing that frequently lead to suboptimal or outright flawed decisions (Korteling et al., 2021). These biases—such as anchoring bias (fixating on initial information), availability bias (overestimating the likelihood of events that are easily recalled), and confirmation bias (favoring information that confirms preexisting beliefs)—are deeply rooted in the structural properties of biological neural networks (Korteling et al., 2021; Tversky & Kahneman, 1974). These limitations stem from the evolutionary mismatch between ancient survival heuristics and modern environments. For instance, the action bias, beneficial for quick decisions in survival scenarios, often results in maladaptive behavior today (Korteling et al., 2021; Baron & Ritov, 2004). This suggests that higher cognitive functions like probabilistic reasoning are evolutionarily recent and remain prone to error, limiting their reliability as a sole foundation for AI systems. Attempts to mitigate these biases through training have largely failed, underscoring the need for AI to go beyond human mimicry and capitalize on alternative forms of intelligence (Korteling et al., 2021).

The distinction between human mimicry and superhuman capabilities forms the core foundation of the MIEF, allowing for a comprehensive framework that evaluates AI systems across dimensions that reflect both human-centric and transhuman potentials.

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Other forms of intelligence, particularly those observed in animals, provide valuable insights for AI development, offering solutions to complex problems beyond human capability. For example, the Bat Algorithm, inspired by the echolocation abilities of bats, has been applied to optimize agricultural irrigation by accurately classifying crop images and detecting water stress. Similarly, the Firefly Algorithm, based on the light-based communication of fireflies, has been instrumental in enhancing irrigation systems, balancing water reservoirs, and improving hydropower management (Wakchaure et al., 2023). These nature-inspired algorithms demonstrate the potential of alternative intelligences to surpass human limitations, showcasing the value of diverse intelligence models in advancing AI innovation. While human intelligence is unparalleled in certain creative and abstract domains, it should not be regarded as the glass ceiling for artificial intelligence. Instead, it represents one piece of a larger mosaic that AI can build upon. By incorporating diverse forms of intelligence, including animal-inspired algorithms and superhuman computational capabilities, AI has the potential to surpass the constraints of human cognition and offer novel solutions to complex problems.

The redefinition of AI as an “alternative intelligence” is pivotal to reshaping its development and societal acceptance. This perspective moves beyond the traditional goal of emulating human cognition and instead sets AI as a complementary force that addresses tasks beyond human reach. For instance, AI-powered drones equipped with real-time mapping and thermal imaging capabilities are deployed in disaster zones to locate survivors and assess structural damage, as highlighted by Papyan et al. (2024). These drones can operate in hazardous conditions, such as thick smoke or unstable debris, where human responders face significant risks. By identifying distress signals like human cries or body heat, these systems demonstrate AI's ability to undertake life-saving operations that humans cannot feasibly execute due to physical and situational constraints. Such examples showcase how redefining AI as an alternative intelligence allows it to transcend human limitations, offering new solutions for critical challenges.

Similarly, in healthcare, AI-powered sepsis prediction models provide another illustration of this redefinition. These systems analyze massive amounts of patient data in real-time, identifying early signs of sepsis with high precision, enabling timely interventions that significantly improve patient outcomes, hours before sepsis symptoms occur (Goh et al., 2021). By integrating predictive algorithms with clinical workflows, such models exceed human capacities in speed and scale of data analysis, effectively addressing the limitations of human decision-making in complex, time-sensitive and high-pressure environments.

Adaptability evaluates an AI system's ability to adjust seamlessly across tasks or dynamic environments. This dimension captures the system's capacity to function effectively in unpredictable or evolving scenarios, a critical requirement for real-world applications. For example, autonomous vehicles exemplify adaptability through their ability to transition between routine driving tasks and reacting to unexpected road conditions, such as detours or weather-related obstacles. The ability to perform consistently in such variable environments reflects an AI system’s potential to address diverse challenges. In the context of the MIEF, adaptability ensures that systems are versatile and robust, maximizing their utility in dynamic and uncertain settings.

Sustainability and ethical alignment are essential dimensions in AI development, ensuring that systems operate efficiently and align with societal priorities. Metrics such as energy consumption per task provide critical insights into an AI system's long-term viability. While large AI models demonstrate groundbreaking capabilities, their high resource demands underscore the need to balance performance with energy efficiency and ecological impact. Incorporating sustainability within the MIEF framework ensures that AI systems remain aligned with global ecological priorities. Ethical alignment emphasizes the importance of transparency, fairness, and accountability in AI decision-making. Autonomous Weapon Systems (AWS) exemplify the tension between technological capability and societal values. While AWS excel in precision and adaptability, they raise profound ethical concerns, such as accountability for lethal decisions and fairness in targeting criteria (Dresp-Langley, 2023). These dilemmas highlight the importance of prioritizing ethical alignment, particularly in high-stakes applications where societal harm may outweigh technical benefits. By addressing these challenges, the MIEF ensures that governance and moral considerations are integral to AI evaluation, promoting responsible and equitable development.

Value Added focuses on AI's unique contributions to advancing human productivity and achieving outcomes beyond human capabilities. For example, in healthcare, AI systems like sepsis prediction models have demonstrated superhuman diagnostic capabilities, predicting critical conditions hours earlier than traditional methods, overall aiding patient care by health-care providers (Goh et al., 2021). Similarly, AlphaFold's protein structure predictions have accelerated scientific discovery, achieving breakthroughs once considered unattainable (Jumper et al., 2021). Beyond specific breakthroughs, AI systems also alleviate burdens on overtaxed human systems, such as reducing physician workloads or optimizing supply chain operations. By addressing inefficiencies and enabling innovations like personalized medicine, the Value Added dimension highlights AI's role as a complement to human efforts rather than a replacement, achieving outcomes previously beyond reach. These examples underscore AI's dual role as a transformative force and an augmentation to human expertise, reinforcing the necessity of multidimensional evaluation frameworks like the MIEF.

Discussion & Conclusions

The Multidimensional Intelligence Evaluation Framework (MIEF) integrates human-centric replication, superhuman capability, adaptability, sustainability, ethical alignment, and value added to address the shortcomings of traditional AI evaluation benchmarks. By uniting these dimensions, the MIEF provides a flexible and balanced framework for assessing AI systems, recognizing that trade-offs are an inevitable aspect of effective evaluation.

Trade-offs underscore the MIEF's adaptability, allowing systems to excel in one dimension while compensating in others. For instance, while AlphaFold lacks broad adaptability, its unparalleled precision in protein folding demonstrates how success in a single dimension can outweigh limitations elsewhere (Jumper et al., 2021). Similarly, autonomous weapon systems (AWS) excel in precision and adaptability in combat but highlight the critical need for ethical oversight to mitigate societal harm (Dresp-Langley, 2023). This holistic approach aligns with Korteling et al.'s argument for redefining intelligence based on outcomes rather than mimicry, enabling AI systems to be assessed for their efficiency and effectiveness (2021).

Human intelligence, while remarkable, is far from infallible. It is limited by cognitive biases, memory constraints, and processing inefficiencies that often lead to suboptimal decision-making. For example, ingrained biases such as anchoring and confirmation bias frequently distort judgment, and higher-order reasoning, such as probabilistic thinking, remains embryonic in evolutionary terms (Korteling et al., 2021; Tversky & Kahneman, 1974). These flaws highlight why human cognition should not be the ultimate benchmark for AI. Instead, the MIEF leverages AI's potential to complement human abilities by excelling in areas where humans are inherently constrained. AI systems like sepsis prediction models exemplify this potential, identifying life-threatening conditions earlier and more accurately than human clinicians can (Goh et al., 2021).

Beyond technical evaluation, the MIEF addresses societal priorities. Sustainability metrics ensure advancements remain viable in resource-limited settings, while ethical alignment fosters trust by prioritizing fairness, transparency, and accountability. Adaptability reflects AI's potential to provide innovative solutions in dynamic and unpredictable environments. Collectively, these dimensions position the MIEF as a forward-thinking framework capable of guiding responsible AI development.

Reevaluating the role of human intelligence within the framework is critical. While human intelligence has inspired AI design, its inherent limitations make it an insufficient standard for evaluating AI systems. Instead, alternative forms of intelligence, including those observed in nature, offer valuable insights for AI development. For instance, algorithms inspired by bat echolocation and firefly communication have provided innovative solutions to optimization problems in agriculture and water management, highlighting the potential of diverse intelligence models to address complex challenges (Wakchaure et al., 2023). By embracing these alternative paradigms, the MIEF enables a broader, more inclusive approach to AI evaluation.

Implementing the MIEF, however, is not without challenges. Defining standardized metrics for abstract dimensions like adaptability and ethical alignment requires collaboration across disciplines, from AI researchers to policymakers. Stakeholder resistance to replacing traditional benchmarks may arise from concerns over practicality or misaligned incentives. Overcoming these barriers necessitates transparent dialogue, pilot programs demonstrating the MIEF's value, and case studies highlighting its application across diverse industries.

Future research should focus on refining metrics and testing the MIEF across real-world contexts. Key questions include: How can adaptability metrics capture the nuances of AI systems across domains? What strategies ensure ethical alignment in high-stakes applications like autonomous systems or medical diagnostics? Exploring hybrid systems that combine human-like and superhuman capabilities offers an opportunity to maximize AI's complementary strengths while maintaining ethical accountability.

In conclusion, the MIEF redefines AI evaluation by integrating human-centric and superhuman dimensions, emphasizing trade-offs as a central principle. By addressing the limitations of traditional benchmarks and recognizing AI's potential to transform society, this framework ensures future systems align with societal values, advancing responsibly and equitably.

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References

Al-kfairy, M., Mustafa, D., Kshetri, N., Insiew, M., & Alfandi, O. (2024). Ethical challenges and solutions of Generative AI: An interdisciplinary perspective. *Informatics*, *11*(3), 58. https://doi.org/10.3390/informatics11030058

Baron, J., & Ritov, I. (2004). Omission bias, individual differences, and normality. *Organizational Behavior and Human Decision Processes*, *94*(2), 74–85. https://doi.org/10.1016/j.obhdp.2004.03.003

Dresp-Langley, B. (2023). The weaponization of artificial intelligence: What the public needs to be aware of. *Frontiers in Artificial Intelligence*, *6*. https://doi.org/10.3389/frai.2023.1154184

Goh, K. H., Wang, L., Yeow, A. Y., Poh, H., Li, K., Yeow, J. J., & Tan, G. Y. (2021). Artificial intelligence in sepsis early prediction and diagnosis using unstructured data in healthcare. *Nature Communications*, *12*(1). https://doi.org/10.1038/s41467-021-20910-4

Jumper, J., Evans, R., Pritzel, A., Green, T., Figurnov, M., Ronneberger, O., Tunyasuvunakool, K., Bates, R., Žídek, A., Potapenko, A., Bridgland, A., Meyer, C., Kohl, S. A., Ballard, A. J., Cowie, A., Romera-Paredes, B., Nikolov, S., Jain, R., Adler, J., … Hassabis, D. (2021). Highly accurate protein structure prediction with alphafold. *Nature*, *596*(7873), 583–589. https://doi.org/10.1038/s41586-021-03819-2

Korteling, J. E., van de Boer-Visschedijk, G. C., Blankendaal, R. A., Boonekamp, R. C., & Eikelboom, A. R. (2021). Human- versus Artificial Intelligence. *Frontiers in Artificial Intelligence*, *4*. https://doi.org/10.3389/frai.2021.622364

Negrotti, M. (1991). Alternative Intelligence. In *Understanding the Artificial: On the Future Shape of Artificial Intelligence* (pp. 55–75). essay, Springer. Retrieved from https://link.springer.com/chapter/10.1007/978-1-4471-1776-6_4.

Papyan, N., Kulhandjian, M., Kulhandjian, H., & Aslanyan, L. (2024). AI-based drone assisted human rescue in Disaster Environments: Challenges and opportunities. *Pattern Recognition and Image Analysis*, *34*(1), 169–186. https://doi.org/10.1134/s1054661824010152

Pease, A., & Colton, S. (2011). On impact and evaluation in computational creativity: A discussion of the Turing test and an alternative proposal. *Proceedings of the AISB Symposium on AI and Philosophy*, *39*, 277–294.

Silva, H. (2024). Alternative intelligence. *Wasafiri*, *39*(3), 80–84. https://doi.org/10.1080/02690055.2024.2349377

Tian, Y., Chen, X., Xiong, H., Li, H., Dai, L., Chen, J., Xing, J., Chen, J., Wu, X., Hu, W., Hu, Y., Huang, T., & Gao, W. (2017). Towards human-like and transhuman perception in AI 2.0: A Review. *Frontiers of Information Technology & Electronic Engineering*, *18*(1), 58–67. https://doi.org/10.1631/fitee.1601804

Tversky, A., & Kahneman, D. (1974). Judgment under uncertainty: Heuristics and Biases. *Science*, *185*(4157), 1124–1131. https://doi.org/10.1126/science.185.4157.1124

Wakchaure, M., Patle, B. K., & Mahindrakar, A. K. (2023). Application of AI techniques and robotics in agriculture: A Review. *Artificial Intelligence in the Life Sciences*, *3*, 100057. https://doi.org/10.1016/j.ailsci.2023.100057